

$$(p^k - p^{k-1})(q^l - q^{l-1})$$

$$= p^k q^l - p^{k-1} q^{l-1} (p+q-1)$$

$$\varphi(n) = \frac{n}{p_1 \cdot p_2 \cdots p_r} \cdot p_1 - 1$$

$$g^0, g^1, g^2, g^3, \dots, g^{p-1}$$

$$(-g)^k \equiv (-1)^k g^k$$

$$-g$$

$$(p-1)! \equiv -1$$

$$\left(\frac{p-1}{2}\right)!$$

$$\left(1 \cdot 2 \cdots \frac{p-1}{2}\right)^2$$

$$\left(1 \cdot 2 \cdots \frac{p-1}{2}\right) (-1)^{\frac{p-1}{2}} \left((-1) \cdot (-2) \cdots \left(-\frac{p-1}{2}\right)\right)$$

$$= (p-1)! (-1)^{\frac{p-1}{2}} = (-1)^{\frac{p+1}{2}}$$